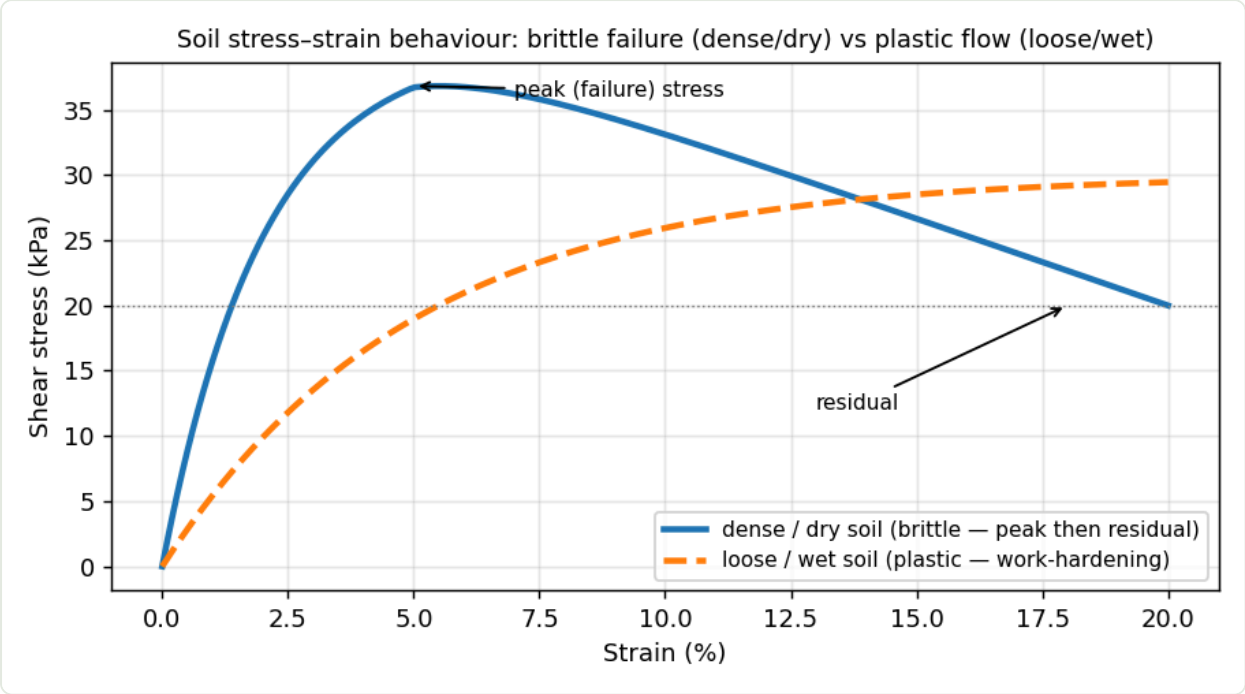
UNIT 6 · TOPIC 2

Stress–Strain Behaviour and Theories of Soil Failure

Syllabus topic: Stress-strain relationships and theories of soil failure

Stress–strain relationships



Soil stress–strain behaviour — brittle failure (dense/dry) vs plastic flow (loose/wet)

Soil is not linearly elastic; its stress–strain curve depends strongly on density and moisture.

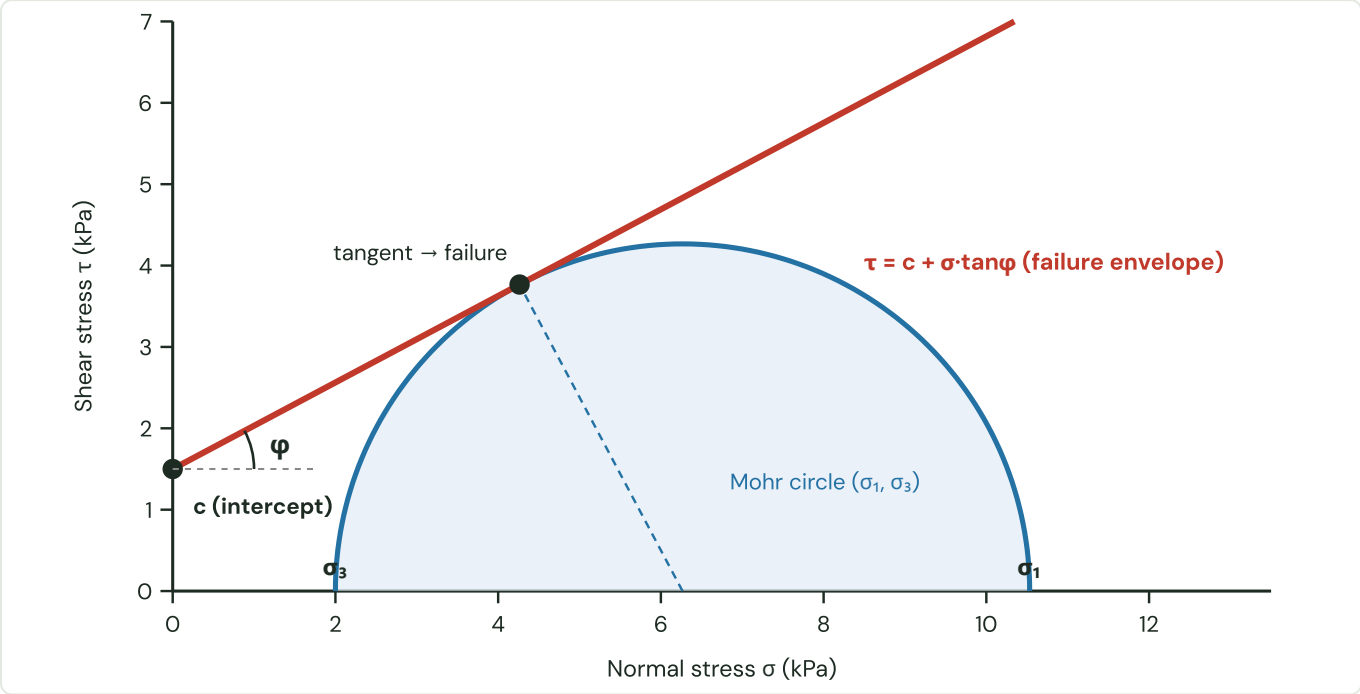
Soil state	Behaviour	Result in tillage
DENSE / DRY	BRITTLE — stress rises to a distinct PEAK (failure), then falls to a lower RESIDUAL as a shear plane forms (strain-softening)	Fails by clean shear/crumbling — good tilth
LOOSE / WET	PLASTIC — stress rises gradually with continued strain (work-hardening); the soil FLOWS/smears rather than shattering	Poor tilth, smearing, compaction

- **Idealised models:** rigid-plastic, elastic-plastic; cohesive plastic soils are treated by the theory of plasticity, brittle soils by limit-equilibrium (Mohr–Coulomb) analysis.
- **Tillage rule:** work soil at the right (friable) moisture — near the PLASTIC LIMIT — for maximum pulverisation at minimum draft. Wet plastic soil smears, clods and compacts; too-dry soil needs excessive energy.
- **Rate effects:** faster loading generally raises apparent strength and draft (dynamic term — Goryachkin, Unit 2, Topic 5).

The Coulomb / Mohr–Coulomb criterion (Liljedahl pp. 240–242)

Coulomb (1776): $\tau = c + \sigma \cdot \tan \varphi$ (σ = normal stress on the failure plane)

- **Mohr** generalised it: failure occurs when the largest Mohr stress circle (principal stresses σ_1, σ_3) becomes TANGENT to this straight-line envelope.
- On a τ–σ plot the envelope's **INTERCEPT is c** and its **SLOPE is φ**.



Mohr–Coulomb failure criterion — example c = 1.5 kPa, φ = 28°, circle drawn at failure (redrawn: the source figure's circle stopped short of the envelope)

Plate/lug derivation — extracting c and φ (pp. 240–242)

Maximum shear force on a plate of area $A = b \cdot l$: $F = A \cdot c + W \cdot \tan\varphi = A(c + \bar{p} \cdot \tan\varphi)$, $\bar{p} = W/A$

1. Shear plates at several normal loads W and record the maximum force F.
2. Plot F against W — a straight line (Fig. 10-2).
3. Read the intercept = **A·c** → c, and the slope = **tan φ** → φ.

- Contact area: track A = b·l (uniform pressure); rubber tyre elliptical footprint A ≈ **0.78·b·l** (Fig. 10-3).
- With c and φ known, the maximum soil THRUST of a traction member is **H = A·c + W·tan φ** — the basis of traction theory (Topic 6).

Special case	Shear strength	Traction consequence
Pure SAND (c = 0)	τ = σ·tan φ	Thrust comes only from weight — add ballast
Saturated CLAY ("φ = 0 condition")	τ = c	Thrust independent of weight — adding weight only causes sinkage and compaction

Liljedahl notes that internal friction φ approaches zero when the surface soil is very wet (p. ~264).

Failure modes and other theories

Soil	Failure pattern ahead of a tool (Gill & Vanden Berg)
Brittle	SHEAR / CRESCENT — forward, sideways and upward rupture into distinct blocks
Plastic	FLOW
Very loose	COMPRESSION

- **Wide blades:** passive earth-pressure / retaining-wall analogy (Rankine/Terzaghi passive resistance).
- **Universal earthmoving (Reece) equation:** draft = soil-weight + cohesion + surcharge + adhesion terms.
- **Narrow tines:** slip-line (plasticity) theory predicts the rupture surface; the CRITICAL DEPTH separates crescent failure (shallow) from lateral compaction (deep) — Topic 3.
- **Compaction as failure by volume change:** under running gear the soil fails by COMPRESSION — particles rearrange into denser packing (Topic 9). The same soil element may shear under a tine yet compact under a wheel, so soil dynamics treats strength AND compressibility together.

Practice MCQs

Q4. The Mohr–Coulomb shear strength equation is:

(a) $\tau = c - \sigma \tan\varphi$ (b) $\tau = c + \sigma \tan\varphi$

(c) $\tau = \sigma - c$ (d) $\tau = c \cdot \sigma$

Q5. In $\tau = c + \sigma \tan\varphi$, the term c represents:

(a) Angle of friction (b) Cohesion (shear strength at zero normal stress)

(c) Normal stress (d) Density

Q6. The angle φ is the:

(a) Slope angle (b) Angle of internal friction

(c) Rake angle (d) Cone apex

Q7. On a τ–σ plot, cohesion c is the:

(a) Slope (b) Intercept on the τ-axis

(c) Area (d) X-intercept

Q8. Mohr's theory says failure occurs when the stress circle:

(a) Is a point (b) Becomes tangent to the failure envelope

(c) Has zero radius (d) Lies below the axis

Q9. For a dry cohesionless sand, cohesion c is:

(a) Very high (b) Zero

(c) Negative (d) Infinite

Q10. In a saturated clay ($\varphi \approx 0$), shear strength τ equals:

(a) $\sigma \tan\varphi$ (b) c (independent of normal stress)

(c) Zero (d) σ

Q11. Adding weight to a drive wheel improves traction in sand because thrust depends on:

(a) Cohesion only (b) Normal load × tanφ

(c) Moisture (d) Colour

Q12. In very wet clay, adding weight to the wheel mainly causes:

(a) More traction
(b) Sinkage and compaction with little traction gain
(c) Cooling
(d) Less slip

Q13. For a plate of area A shearing soil, maximum force F =

(a) A/c
(b) Ac + W tanφ
(c) W/A
(d) c − W

Q14. Plotting maximum shear force F versus normal force W gives a line whose slope is:

(a) c
(b) tanφ
(c) A
(d) W

Q15. The intercept of that F–W line equals:

(a) tanφ
(b) A·c
(c) W
(d) φ

Q16. The contact 'footprint' area of a rubber tyre is approximated as:

(a) b·l
(b) 0.78 b·l
(c) $\pi b^2/4$
(d) 2bl

Q25. Dense, dry soil shows a stress–strain curve that is:

(a) Purely linear
(b) Brittle — a peak then a lower residual
(c) Flat
(d) Ever-rising with no peak

Q26. Loose, wet soil tends to:

(a) Shatter cleanly
(b) Flow/smear (plastic, work-hardening)
(c) Stay rigid
(d) Explode

Q27. Best pulverization at least draft is obtained by tilling soil near its:

(a) Liquid limit
(b) Friable moisture (near plastic limit)
(c) Saturation
(d) Oven-dry state

Q28. Tilling wet plastic soil causes:

(a) Fine tilth
(b) Smearing, clods and compaction
(c) Lower draft
(d) No effect

Q33. Under a wheel, soil often fails by:

(a) Shear only
(b) Compression (volume change / densification)
(c) Melting
(d) Evaporation

Q34. Ahead of a narrow tine, brittle soil typically fails by:

(a) Flow
(b) Crescent/shear rupture into blocks
(c) Compression only
(d) No failure

Q35. The universal earthmoving (Reece) equation expresses tool draft as a sum of terms for:

(a) Only weight
(b) Soil weight, cohesion, surcharge and adhesion
(c) Fuel and speed
(d) Slip and sinkage

Answers

Q4 — (b) $\tau = c + \sigma \tan\varphi$ — *Liljedahl p. 241 — $F = Ac + W\tan\varphi \Rightarrow \tau = c + \sigma\tan\varphi$.*

Q5 — (b) Cohesion (shear strength at zero normal stress)

Q6 — (b) Angle of internal friction

Q7 — (b) Intercept on the τ-axis — *c is the τ-intercept, φ the slope.*

Q8 — (b) Becomes tangent to the failure envelope — *Mohr circle tangent to the envelope = failure.*

Q9 — (b) Zero — *Sand $c \approx 0$ — weight-driven thrust.*

Q10 — (b) c (independent of normal stress) — *Saturated clay $\varphi \approx 0$ so $\tau = c$ — weight-independent.*

Q11 — (b) Normal load × tanφ

Q12 — (b) Sinkage and compaction with little traction gain — *Liljedahl notes φ → 0 as soil gets very wet; ballast helps in sand, not in wet clay.*

Q13 — (b) Ac + W tanφ — *Plate/lug derivation, Liljedahl pp. 240–242.*

Q14 — (b) tanφ — *Fig. 10-2 — the F–W plot gives slope tanφ.*

Q15 — (b) A·c — *Fig. 10-2 — intercept A·c.*

Q16 — (b) 0.78 b·l — *Fig. 10-3 — tyre footprint A ≈ 0.78 b·l.*

Q25 — (b) Brittle — a peak then a lower residual — *Brittle vs plastic stress–strain — the friable-moisture tillage rule follows.*

Q26 — (b) Flow/smear (plastic, work-hardening)

Q27 — (b) Friable moisture (near plastic limit) — *Least draft and best pulverisation near the plastic limit.*

Q28 — (b) Smearing, clods and compaction

Q33 — (b) Compression (volume change / densification)

Q34 — (b) Crescent/shear rupture into blocks

Q35 — (b) Soil weight, cohesion, surcharge and adhesion

Sources: Gill, W. R. & Vanden Berg, G. E. — Soil Dynamics in Tillage and Traction, USDA Agriculture Handbook No. 316 (1967); Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 10, pp. 240–252; Kepner, Bainer & Barger, Ch. 6; ASAE S313.2 (cone penetrometer); classical soil mechanics (Coulomb 1776; Mohr); NPTEL soil-dynamics lectures.